THE STUDY AND IMPLEMENTATION OF REAL-TIME FACE RECOGNITION AND TRACKING SYSTEM

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Abstract:

During the past several years, face recognition in video has received significant attention. For the video monitoring and artificial vision, real time face recognition has very important meaning. The current method is still very susceptible to the illumination condition, non-real time and very common to fail to track the target face especially when partly covered or moving fast. In this paper, we propose to use AdaBoost Cascade for face detection and then in order to recognize the candidate faces, they will be analyzed by the hybrid Wavelet and LDA method. After that, Mean shift will be invoked to track the face. The implementation shows that the algorithm has quite good performance in terms of real-time and the tracking procedure is triggered accurately.

Keywords:

Face recognition; LDA; AdaBoost; Mean shift

1. Introduction

PTZ cameras are extensively used for surveillance, the environments, either wide-area or small region, usually requires multi-cameras to active in work. As a rule these cameras operates under a control system. In a single camera environment, we apply the motion detection algorithm to detect a new person intruding the surveillance area, and then start tracking procedure up. Beside these algorithms we require the data fusion process for multi cameras environments. Whether these cameras launch the tracking procedure for moving objects or not? When does the control system begin to effect or record? If these decisions is correct, we can save process time, storage space, and increasing the system efficiency. For people Surveillance, these correct decisions usually depend on face recognition since it is one of the most attractive biometrics[1]. Furthermore, it can be acquired without the cooperation or knowledge of the subject. Subsequent procedure will be easy to progress as soon as we identify this object. However, machine recognition of faces is a challenging problem. The appearance of a face is significantly susceptible by illumination, pose and facial expressions. Many face recognition from still images has been investigated. One of the well-known algorithms in this category is the eigenfaces[2] which projects the facial images to PCA subspace and performs matching on a subset of the most significant eigenvectors. However, it requires perfect normalization of the faces and is sensitive to variation in illumination, pose and expressions. Another algorithm is Fisherfaces[3], which performs Linear Discriminant Analysis (LDA) for face recognition and requires multiple instances per face e.g. under varying illumination, pose or expressions. This method maximizes the ratio of between-class variance to the within-class variance in any particular data set thereby guaranteeing maximal separability. It assumes that the classes are linearly separable which is not always true in the case of faces. Moreover, within-class variation due to changing illumination and pose can cause more variation in faces compared to within-class variations. The prime difference between LDA and PCA is that PCA does more of feature classification and LDA does data classification. Those Methods which extract features from the face as a whole, like eigenfaces and fisherfaces, are called holistic methods[4]. On the other hand, algorithms which extract local features, such as the eyes and nose, and match their relative locations or statistics are called feature-based methods[4]. An examples of feature-based methods is the elastic bunch graph matching[5]. Compared to holistic methods, feature-based methods require higher resolution facial images to extract good quality local features. Face recognition algorithms that extract both holistic features and local features are called hybrid methods[4].

Compared to still images, surveillance videos contain more variation because of the additional temporal dimension. Movement velocity and position of the monitored object has relation with this dimension. Resolution of the subject, image definition and image noise also varies with this dimension. Moreover, surveillance cameras have a wide field of view

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because they are required to survey large areas. Surveillance cameras are also meant to monitor activities rather than just focus on faces. This makes video-based face recognition extremely tough in real time. It is very useful that the system can detect, recognize face and then tracking subject in real time. However, we will be encounter several currently problems which like false retrieval of faces, the influence of image noise, the low accurate rate of face recognition, the lack of real time and lose track of the target. In this paper, we addressed these problems by first using the cascade AdaBoost[6] to detect the face. After that, face candidate region will be analyzed by the algorithm based on the combination of discrete digital wavelet, LDA. According to the similarity of the candidate face judge the target who be allowed in this area or not. If not, then start tracking procedure up and the trail record on the control system. The results of the experiment indicate that the proposed method produces a significant improvement which includes a substantial in error rate and in time of recognizing and tracking. The rest of the paper is organized as follows. Section 2 provides the system framework. Section 3 details about related work which composed of AdaBoost cascade for face detection and wavelet, LDA for face recognition .describes the algorithm of tracking object. The user interface is presented in Section 4. Section 5 gives the conclusion.

2. System framework

A PTZ camera tracking surveillance framework which includes face recognition algorithm and camera tacking algorithm is proposed in this paper. The proposed framework uses a PTZ camera to undertake the task of monitor and tracking. In this paper, we consider a scenario of monitoring the human entering through a single door into an enclosed rectangular environment. Fig. 1 shows the software and hardware architecture of the PTZ Camera based real-time surveillance system. The infrastructure of our system is composed of several software and hardware. Our infrastructure is included the PTZ camera in indoor places, a network video recorder (NVR), a database server that stores information about the video, a monitor control server shows the summary, video and location analysis components, and user interface component. The NVR receives video stream from the PTZ camera. The video analysis component retrieve live video stream from the NVR, and analyzes video information and stores results in the database. The location component retrieves object information and estimates the position of the object. Finally, user interface component display information.

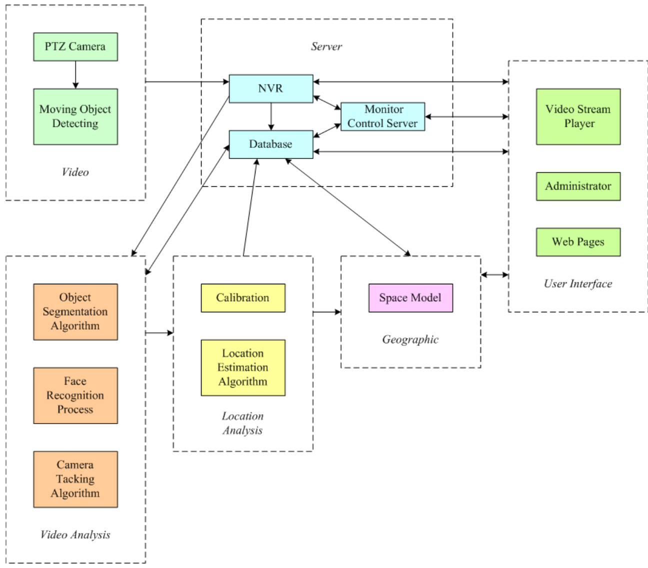


Figure 1. The framework of real-time surveillance system.

A surveillance framework which adopts a cooperative strategy is proposed, with the combination of camera rotation algorithm and face recognition algorithm. The aim of the surveillance framework is to recognize invader and carry out the tracking task, when invader has been detected. The proposed framework uses several cameras to undertake the task of monitoring the door to detect any new intruders while tracking a current intruder. A diagrammatic representation of the surveyed premises used for experiments in our framework has been shown in Fig. 2. The proposed framework uses PTZ cameras to undertake the task of monitor and track. There are total eight cameras in this example. The AXIX 215 camera was used for implementation. The camera accepts the HTTP API comments for requesting motion video, still images, setting and inquiry of camera parameters and control commands for panning, tilting, zooming and focusing the camera. One of these cameras constantly monitors the entrance to detect new intruders (camera 7 in Fig.2). This monitor camera is stable and always focuses on the entering door. One of the other cameras, called track camera focuses on the intruder to obtain quality images. In this system, we desire to continuously track as target as possible with multiple active cameras subject to limited fields of view. If the object has discovered from entrance at the current time instant, the multi-camera surveillance systems will determine which camera should track and rotate in order to focus the target at the next time instant.

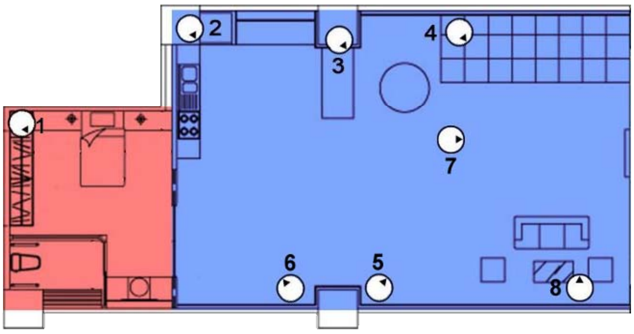


Figure 2. The surveyed premises used in our framework.

3. Real-time surveillance system in real work

We implement a real-time surveillance system consist face recognition and tracking task in real work. Figure 3Figure shows the block diagram of face recognition and tracking system. The camera detects the face as someone approaches the front door, and appears in its field of view. After Image pre-processing and LDA transformation, camera recognize the object is one of family members. Otherwise, system will automatically send a warning to default users, and notify indoor cameras to track the object continuously while updating its pan, tile and zoom. Indoor cameras stop tracking and set to home position, only when users reply that it is an innocent person. Some of the important components of Figure 3 are detailed as follow.

3.1. Face detection

We implement the real-time surveillance system which consist face recognition and invader tracking via OpenCV libraries[7]. OpenCV provides a set of open source programming libraries for computer vision applications. The system uses the socket to send HTTP API commands to the camera and retrieve MJPG streaming. Each frame separate to a JPG frame, and then store it as the IplImage data structure (transform it into the IplImage data structure). The OpenCV built in object detector which is based on Haar-like features was used to detect faces[8-9]. Haar-like features are widely used in face searching have been trained to accurately represent human faces. In a Haar-like feature approach, feature values are obtained by summing up the values of pixels in each region of a face image and weighting and then summing up the regional sums, instead of directly using the values of the pixels of the face image.

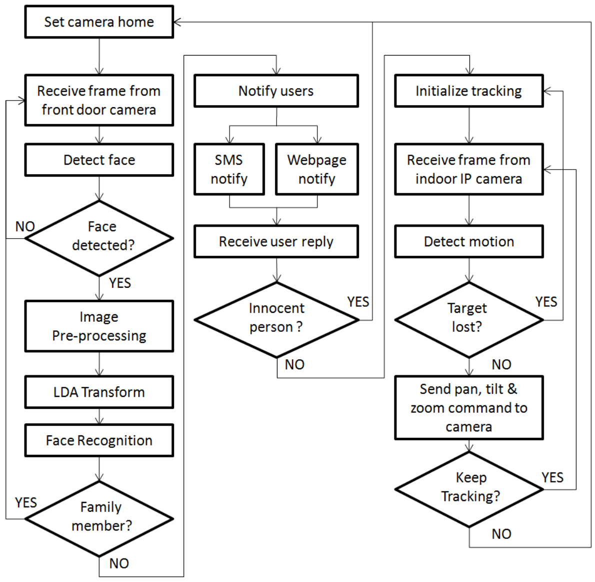


Figure 3. Block diagram of real-time surveillance system.

The cascade Adaboost classifiers which is based on using haar-like feature is a good and robust method for characterization and detection approach[6]. OpenCV provide several already trained classifiers for face detection. We use the face detection classifier through OpenCV libraries in our implementation. Figure 4 is shown the result.

Figure 4. Face detection.

3.2. Face recognition

Before the face recognition, we must perform the image pre-processing task first. The image pre-processing is important to whole task, which can improve the face recognition accuracy rate and image analysis. Figure 5 shows the step of image pre-processing. The size of human face will be normalized to 92 pixels × 112 pixels. Gray-level transformation reduces the impact of unconstrained illumination problem, and gets lower computation. Furthermore, histogram equalization makes the pixels in face image have the same intensity by the transfer function. Discrete wavelet transform filters out high-frequency noise and reduces image dimension. The resulting images relative to the block of Figure 5 respectively as shown in Figure 6.

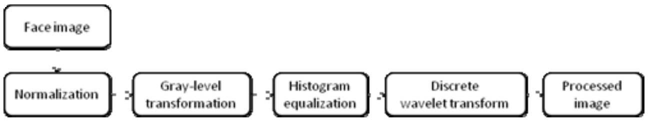


Figure 5. Block diagram of image pre-processing.

Figure 6. The resulting product of image pre-processing.

LDA is one of the most popular linear projection techniques for feature extraction. LDA is a statistical approach for classifying samples of unknown classes based on training samples with known classes. This technique aims to maximize between-class variance and minimize within-class variance. (Each block represents a class, there are large variance between classes, but little variance within classes.) Suppose there are C classes. Let c_k be the mean vector of k class, N_k be the number of samples within class k and c is the average value of all samples in the database. The within-class scatter matrix S_W and the between-class scatter matrix S_B are defined as below:

$$S_W = \sum_{k=1}^{C} \sum_{y_{kj} \in y_k} (y_{kj} - c_k)(y_{kj} - c_k)^T \quad (1)$$

$$S_B = \sum_{k=1}^{C} N_k (c_k - c)(c_k - c)^T \quad (2)$$

LDA computes a transformation that maximizes the between-class scatter while minimizing the within-class scatter. That is, to find a project matrix W that maximizes the ratio of between-class scatter against within-class scatter. We hope to obtain the optimal W_{opt}.

$$W_{opt} = \arg\max \frac{|W^T S_B W|}{|W^T S_W W|} \quad (3)$$

It can accomplish classify samples, while we transform the image to the feature space as well as lower the impact by the environment simultaneously.

Figure 7 shows the block diagram of face recognition system. After face detection, image pre-processing and LDA transformation, the test image and training data have to be compared until finding a matched class. If the test image can't find a pair match, then bring the tracking task into play.

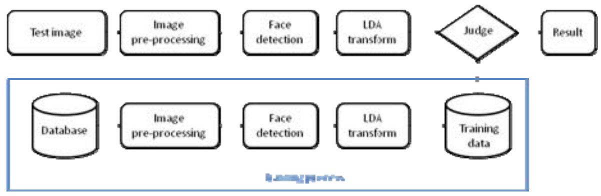


Figure 7 Block of face recognition.

3.3. Camera tracking task

In any tracking system, segmenting out humans from the background is necessary prior to tracking human in the camera scenes. The detection of moving objects in video sequences is an important and difficult research issue in tracking applications. One commonly used approach to foreground extraction is to subtract out the background from the video frames. It consists in maintaining a reference background image for the scene observed and extracting new foreground objects by comparing the current image against the reference background.

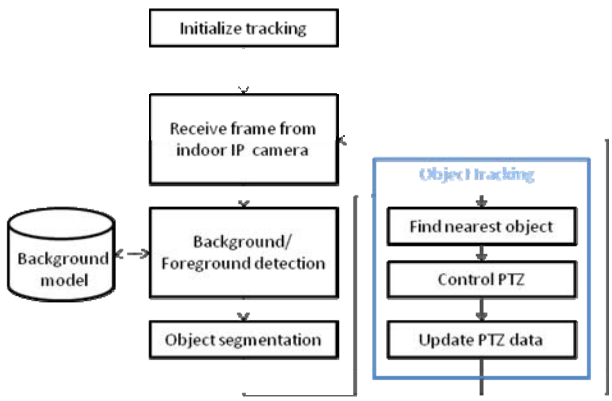


Figure 8 Block diagram of tracking procedure.

Background/Foreground detection, Object segmentation and Object tracking are three component of tracking system. Figure 8 shows the block diagram of tracking system. Adaptive background/foreground detection is used to find the moving object with pre-created background. After that, cut the foreground object and track it by control the indoor PTZ camera. The algorithm of adaptive B/F detection is described as below.

1. $d_i = | f_i - \mu |$
2. If $d_i > T$, f_i belong to foreground; otherwise, it belong to background.

Where f_i is a pixel in a current frame i. μ is a pixel of background model. d_i is absolute difference between f_i and μ. b_i B/F mask-0: background. 0xff: foreground. T is threshold. α is learning rate of the background. The block diagram of this approach shows in Figure 9. On the part of tracking object, many algorithms had been proposed [10-14], but still popular issue. In this paper, the step of object tracking has implemented by using mean shift algorithm. The OpenCV library provides the mean shift source program. Mean shift algorithm is nonparametric statistical method, which can make rapidly optimal matching during object tracking. So it is suitable for real-time tracking system. Therefore, we use the algorithm in our tracking system.

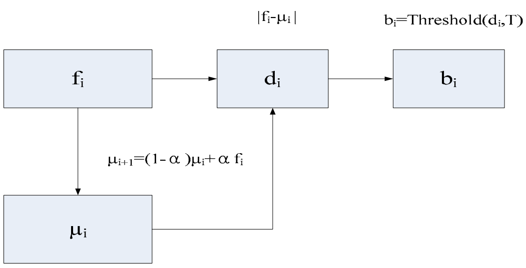


Figure 9. Block diagram of the B/F detection.

4. User interface

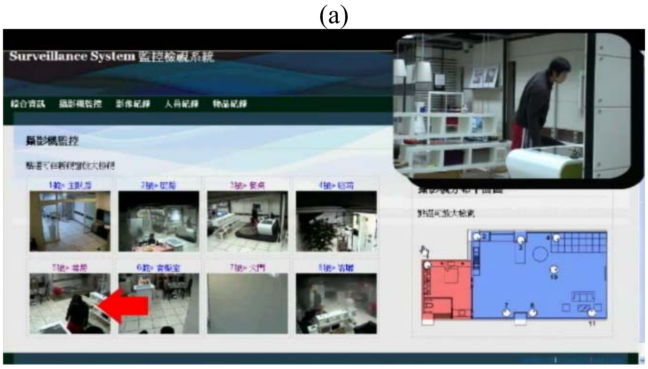
We implement web user interface component display current information and records. The web interface has the characteristic of cross-platform. Users connect the web interface to obtain the information anytime anywhere. The information includes current parameters and states. Figure 10 shows the comprehensive information consist of all information. We can observe the camera direction, face recognition result, variation of event, etc. the variation of event can be adjusted through the trajectory. The camera display has shown in Figure 11. This page shows the all of the current camera view. We can rotate the camera to any angle via pan, title zoom control. Besides, this page supports the manual tracking control. User marks any frames which under the camera view to start/close the tracking task. The face recognitions results, which as shown in Figure 12, shows the invader's face and recognition results. The user can discriminate the identity of invader by itself. The Figure 13 shows the invader tracking task process. When the invader has been detected, we start the tracking task up. This task can record all of the behavior of invader to MJPEG frames and these frames will be pile up.

Figure 10. Comprehensive information.

Figure 11. Camera display.

Figure 12. Face recognition results.

(a)



(b)

Figure 13. invader tracking.

5. Conclusions

In this paper, we implement a real-time surveillance consist of face recognition and invader tracking which based on OpenCV. OpenCV provides a set of open source programming libraries which suitable for real-time surveillance system. In this paper, we propose to use AdaBoost Cascade combined with skin model for face detection and then in order to recognize the candidate faces, they will be analyzed by the hybrid Wavelet and LDA method. After that, Mean shift and Kalman filter will be invoked to track the face. The user interface provides the current state, camera view and records for cross-platform user. The experimental results show that the algorithm has quite good performance in terms of real-time and accuracy.

Acknowledgment

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